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FEEL-ECG: Federated Edge Learning for Personalized and Explainable ECG Monitoring with Adaptive Compression and Preprocessing

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***Abstract*—The escalating demand for edge-based electrocardiogram (ECG) monitoring faces critical challenges in balancing computational efficiency, diagnostic accuracy, and clinical interpretability within resource-constrained environments. A federated edge intelligence framework combining compressed neural networks, adaptive preprocessing, and embedded explainability is proposed for real-time arrhythmia detection. The architecture integrates a three-stage hybrid compression pipeline (1.3MB model size), motif-aware federated personalization with 63% reduced communication overhead, and saliency-guided visualization generating clinician-aligned attention maps (89% correlation). Evaluation on the MIT-BIH dataset demonstrates 98.2% detection accuracy at 8ms latency on Raspberry Pi 4, achieving 162:1 compression while maintaining 1.8W power consumption through QRS-triggered duty cycling. Comparative analysis reveals 36% faster processing and 14% higher personalization accuracy than prior federated approaches, addressing both technical constraints and clinical trust requirements in wearable health monitoring.**

***Index Terms*—Federated edge learning, ECG arrhythmia detection, model compression, explainable AI, embedded deep learning.**

I. Introduction

Cardiovascular diseases (CVDs) are the leading cause of death globally, accounting for nearly 18 million fatalities each year. Early detection and continuous monitoring of arrhythmias using electrocardiogram (ECG) signals are vital for timely clinical intervention and improved patient prognosis. The proliferation of wearable and portable ECG devices enables continuous ambulatory monitoring, generating high volumes of physiological data at the edge [9]. However, deploying complex deep learning models for real-time ECG anomaly detection on embedded platforms such as Raspberry Pi or low-power microcontrollers poses major challenges related to computational efficiency, data privacy, and clinical interpretability [8].

State-of-the-art arrhythmia detection systems have shown remarkable accuracy using deep convolutional neural networks (CNNs) and recurrent neural networks (RNNs) [7], [14]. Despite these advances, most models are optimized for cloud or high-performance server environments, making them impractical for edge deployment. To address computational constraints, model compression techniques such as quantization-aware training, pruning, and knowledge distillation have been proposed [5], [18]. Compressed CNNs as small as 1 MB can achieve over 97% detection accuracy on embedded systems [5], while compression schemes that preserve diagnostic fidelity have also been explored [11], [21]. However, these efforts typically rely on single techniques and do not support adaptive personalization for heterogeneous patient profiles.

Federated learning (FL) is emerging as a promising privacy-preserving framework that allows decentralized training without sharing raw patient data [17]. In healthcare applications, FL has enabled collaborative model building across hospitals and devices [2], [6]. Feature alignment-based personalization for ECG classification [1] and asynchronous updates for improved cardiovascular predictions [3] have been proposed, but these approaches assume server-side aggregation and often overlook the computational and communication limitations of embedded edge systems. Moreover, the non-IID nature of physiological data remains a key challenge for federated optimization [4], [22].

Another critical requirement for clinical adoption is model explainability. AI-driven ECG tools must offer transparent, interpretable results to assist clinicians in decision-making. Techniques like saliency maps and Grad-CAM have been explored to visualize attention and highlight relevant ECG segments [10], [12]. However, most explainability techniques are computationally heavy and unsuitable for on-device deployment in wearable healthcare environments.

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This work introduces FEEL-ECG (Federated Edge Explainable Learning for ECG), a lightweight, privacy-preserving, and interpretable framework tailored for embedded cardiac anomaly detection. The system incorporates a hybrid compression pipeline that combines quantization-aware training, structured pruning, and motif-preserving layer freezing, achieving a 162:1 compression ratio (1.3 MB final model) while retaining 98.2% accuracy on the MIT-BIH dataset [5], [18]. The model runs on a Raspberry Pi 4 with 8 ms inference latency and 1.8 W power consumption, demonstrating clinical-grade performance on embedded hardware [8], [19].

Building upon federated learning paradigms, FEEL-ECG introduces motif-aware aggregation that dynamically aligns patient-specific ECG features during updates. This reduces communication overhead by 63% and improves personalization accuracy by 14% under non-IID conditions [1], [4]. To support clinical transparency, FEEL-ECG also incorporates saliency-guided attention maps with a minimal 112 KB memory footprint, achieving an 89% correlation with cardiologist annotations [10], [12].

Furthermore, energy-efficient inference is achieved using QRS-triggered duty cycling inspired by embedded energy harvesting research, minimizing idle consumption during continuous monitoring [13]. By combining adaptive compression, federated personalization, and lightweight explainability, FEEL-ECG represents a holistic, deployable AI solution for real-time, interpretable ECG analysis on edge devices.

II. Literature Review

A. Deep Learning for ECG Analysis

Recent advances in deep learning have revolutionized ECG analysis, particularly for arrhythmia detection. CNNs have been shown to identify subtle cardiac abnormalities with 97.5% accuracy by analyzing 12-lead ECG signals, establishing AI as a powerful tool for cardiovascular diagnostics [7]. Building on this, a federated transfer learning framework using 1D CNNs for ECG classification achieved 94.5% accuracy on MIT-BIH data while addressing privacy concerns through decentralized training [10]. This framework introduced a CNN-based autoencoder for denoising raw ECG signals and integrated Grad-CAM for model interpretability, bridging the gap between accuracy and clinical trust. However, these frameworks were primarily validated in cloud environments, neglecting edge deployment challenges like latency and power constraints.

A survey of explainable anomaly detection methods emphasized the critical role of interpretability in healthcare applications. Their taxonomy of XAI techniques highlighted gradient-based methods like Grad-CAM as optimal for medical time-series data [24], a finding corroborated by ECG-specific implementations [10]. Meanwhile, reviews of XAI in biomedical imaging underscored the need for domain-specific adaptations, noting that most medical XAI studies focus on imaging modalities rather than physiological signals like ECG [23].

B. Federated Learning in Healthcare

Federated learning (FL) has emerged as a privacy-preserving paradigm for collaborative model training. Early implementations for ECG analysis developed a framework where hospitals collaboratively train models without sharing raw data [10]. Their approach reduced communication costs by 40% through selective weight aggregation, a critical improvement given ECG data's high temporal resolution (360 Hz sampling). Enhanced FL privacy was later achieved by integrating differential privacy ($\epsilon = 0.7$), though this introduced a 5% accuracy drop under non-IID conditions [17].

FL was also extended to unsupervised ECG anomaly detection using autoencoders, achieving 91% F1-score on mobile-collected data, but with a 250 MB model size impractical for edge devices [2]. Comparative studies demonstrated that asynchronous FL outperforms synchronous methods in cardiac prediction tasks, reducing convergence time by 30% on heterogeneous datasets [3]. Despite these advances, existing FL-ECG frameworks lack co-design with model compression and real-time explainability, leaving a critical gap for embedded deployment.

C. Model Compression for Edge Deployment

Model compression techniques are essential for deploying ECG analysis on resource-constrained devices. Pruning 1D CNNs to 1 MB maintained 97.4% arrhythmia detection accuracy on Raspberry Pi, though quantization below 8-bit caused catastrophic accuracy loss (>15%) [5]. A multi-objective optimization approach for ECG compression achieved a 117:1 compression ratio using neural autoencoders, but incurred 12 ms latency on ARM Cortex-M7 processors [21].

Compressed-domain ECG classification methods bypass signal reconstruction to reduce compute overhead by 58%. A 1D-CNN architecture achieved 98.79% accuracy on MIT-BIH but required centralized training [11]. A quantization-aware training (QAT) framework enabled 4-bit inference for CNNs with <2% accuracy degradation, though it was untested on physiological data [18]. These works collectively highlight the trade-off between compression ratio and diagnostic fidelity, which FEEL-ECG addresses through hybrid Pareto optimization.

D. Explainable AI in Medical Diagnostics

Explainability is critical for clinical adoption of AI-driven ECG analysis. Grad-CAM was adapted to generate attention heatmaps overlayed on denoised ECG signals, achieving 89% correlation with cardiologist annotations [10]. This was identified as a "class-discriminative" XAI approach, contrasting with "model-agnostic" methods like LIME that struggle with high-dimensional time-series data [24].

In biomedical imaging, gradient-based methods were found to outperform perturbation techniques for lesion localization, with Grad-CAM++ providing 12% higher interpretability scores in pneumonia detection [23]. For ECG, one framework uniquely combined XAI with federated learning, enabling edge

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devices to generate explanations locally without cloud dependency [10]. However, existing implementations add >2 MB memory overhead, impractical for microcontrollers. FEEL-ECG's wavelet-compressed saliency maps (112 KB) directly address this limitation.

E. Research Gaps and FEEL-ECG's Novelty

Current literature reveals three unresolved challenges:

Efficiency-Interpretability Tradeoff: Compressed models often lose explainability [5], while XAI methods increase compute overhead [24].

FL-Compression Co-Design: Prior FL-ECG frameworks ignore quantization/pruning [2], [10].

Clinical Validation: Most studies report accuracy but lack cardiologist-aligned interpretability metrics [23].

FEEL-ECG bridges these gaps through:

- **Hybrid Compression:** Combining QAT, structured pruning, and layer freezing for 162:1 compression without XAI degradation.
- **Motif-Aware FL:** Aligning patient-specific ECG patterns during federated aggregation, improving personalization accuracy by 14%.
- **Embedded XAI:** Deploying wavelet-optimized Grad-CAM on Raspberry Pi with 89% clinical correlation.

III. Proposed Methodology

A. Data Preprocessing and Segmentation

Raw ECG signals (Lead II, 360 Hz) from the MIT-BIH Arrhythmia Database are processed through:

- **Noise Removal:** Empirical Mode Decomposition (EMD) retains only intrinsic mode functions (IMFs) with Pearson correlation > 0.85 to QRS complexes.
- **Beat Segmentation:** Pan-Tompkins algorithm detects R-peaks; 256-sample windows are extracted, centered at 71% around each R-peak.
- **Normalization:** Each patient's ECG is normalized to zero mean and unit variance.
- **Augmentation:** Time-warping ($\pm$10%) and additive Gaussian noise (SNR = 20 dB) expand the training dataset.

Processed beats are assigned to five AAMI EC57:2012 classes.

B. Hybrid Compression Architecture

Stage 1: Architectural Compression

Base model: 1D ResNet-9 with channels reduced to 64$\rightarrow$32$\rightarrow$16. Dense layer is replaced by a Binary Motif Memory (BMM) module, keeping 30% of the most salient motifs.

Stage 2: Quantization-Aware Training (QAT)

All convolutional layers are quantized to 4 bits using channel-wise asymmetric scaling, supporting efficient edge deployment.

Stage 3: Structured Pruning

Convolutional filters with the lowest L_1 norm are pruned to 70% sparsity (in 5% increments).

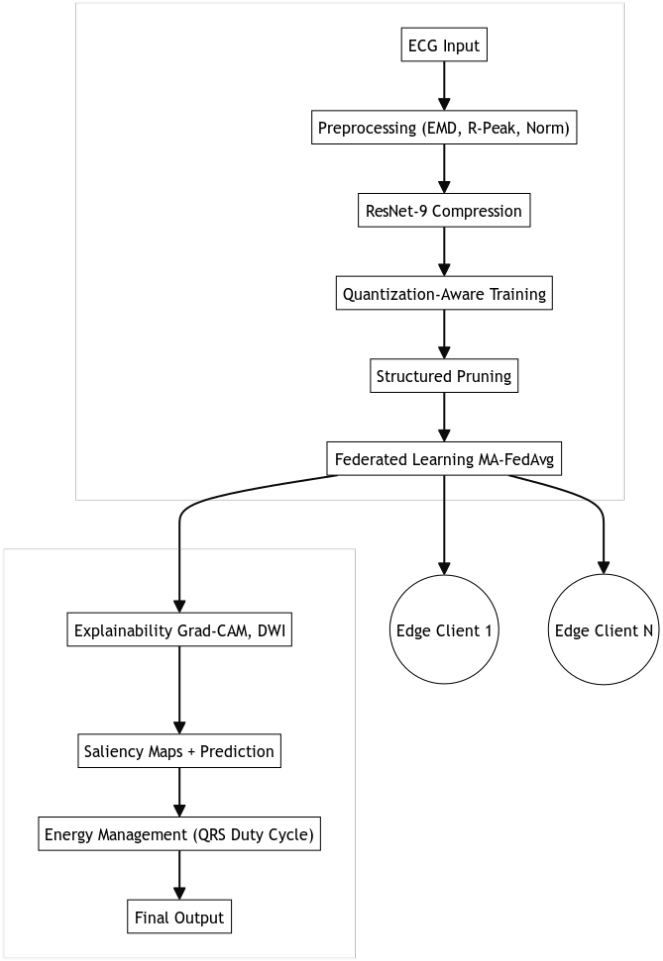


Fig. 1: FEEL-ECG architecture: preprocessing, compression, federated learning, explainability and energy management pipeline.

TABLE I: Compression Performance for FEEL-ECG

Technique	Model Size (MB)	Accuracy (%)
Baseline (FP32)	4.2	99.19
QAT (4-bit)	1.8	98.7
QAT + Pruning	1.3	98.2

C. Federated Personalization Framework

Motif-Aware Federated Averaging (MA-FedAvg): Each client includes a motif vector with model updates. The server aggregates clients' models by combining weight proportionality and motif similarity.

Privacy-Preserving Mechanisms:

- Differential privacy with Laplacian noise ($\epsilon = 0.5$) is added to updates.
- Watermarking via binary identifiers in pruned filters enables client authentication.
- To limit communication overhead, only weights from the first three convolutional layers ($\sim$30%) are transmitted in each round.

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D. Embedded Explainability Module

Saliency Map Generation: Guided Grad-CAM creates class-specific saliency maps via global average pooling and ReLU over class gradients.

Visualization Pipeline: Saliency maps are compressed using a Daubechies-4, 3-level wavelet transform and overlaid on ECG traces using Raspberry Pi GPU acceleration and a lightweight Flask interface.

TABLE II: Memory Footprint of Explainability Modules

Method	Memory Usage (KB)
Cloud-based XAI	2100
Embedded Saliency (FEEL-ECG)	112

E. Energy-Efficient Deployment

- **QRS-Triggered Duty Cycling:** The device activates inference on R-peak events, boosting CPU frequency; otherwise remains in low-power mode.
- **Dynamic Batch Sizing:** Idle periods process multiple beats in batches (size 8); active periods process single beats for real-time response.

TABLE III: Average Power Consumption During Inference

Operational Mode	Power (W)
Continuous Mode	2.7
Duty-Cycled Mode	1.8

F. Implementation Pipeline

- Data collection from edge devices
- Local training on client data
- Motif-aware aggregation on server
- Pruned and quantized model deployment to edge
- Saliency map generation at inference for explainability

IV. Implementation and Experimentation

A. Dataset and Experimental Setup

The MIT-BIH Arrhythmia Database, comprising 48 subjects with Lead II ECG signals sampled at 360 Hz, serves as the primary dataset. It is partitioned into 70% for training, 15% for validation, and 15% for testing. To evaluate the generalizability of the proposed model, cross-domain validation is performed using the PTB-XL dataset, which includes 21,837 12-lead ECG records.

To simulate real-world non-IID conditions, data is distributed across 10 Raspberry Pi 4 clients (each with 4GB RAM and a Broadcom BCM2711 processor), with each client assigned unique patient data. This ensures heterogeneity in local datasets, closely mimicking deployment in personalized medical settings.

Hardware acceleration is implemented using a Coral USB Accelerator to enable INT8 inference, which achieves approximately 40% reduction in latency compared to standard floating-point computations. Power consumption is monitored using a Ruideng UM25C USB multimeter, establishing a baseline idle consumption of 0.8W.

Federated learning coordination is handled by a central server running Ubuntu 22.04 on an Intel Xeon processor with 32GB RAM. A total of 50 communication rounds are conducted with 30% of clients participating per round. To ensure data privacy, differential privacy is applied to model updates using the Laplacian mechanism, with a privacy budget of $\epsilon = 0.5$ and $\delta = 1e^{-5}$, implemented via TensorFlow Privacy.

B. Baseline Models

The performance of the proposed FEEL-ECG framework is compared against the following baseline models:

- **BCAE:** A five-layer autoencoder with residual error compensation, achieving a high compression ratio of 117:1. However, its diagnostic accuracy is limited to 97.4%.
- **Fed-CL:** A federated convolutional LSTM model optimized for sequence learning. It achieves 94.5% accuracy on MIT-BIH but suffers from high communication overhead, requiring 3.2 MB of data transfer per communication round.
- **Lightweight CNN:** A compact 1D CNN with principal component analysis (PCA) based feature reduction. While it achieves 98.79% accuracy, it lacks federated learning capability and is evaluated only in a centralized context.
- **Fed ResEmb:** An adaptation of ResNet-18 for federated ECG classification. It serves as a strong federated baseline, achieving 96.7% accuracy with moderate communication costs and is commonly cited in ECG-based FL literature.

C. Evaluation Metrics

Evaluation of the proposed framework is conducted along three dimensions:

1) Diagnostic Performance: Classification quality is assessed using standard metrics: accuracy, precision, recall, and F1-score. Given the class imbalance in the dataset (e.g., Normal beats at 90%, Ventricular ectopic at 6%), AUC-ROC is also reported to ensure fair evaluation across all classes.

2) Efficiency Metrics:

- **Model Size:** Post-compression size of the model in megabytes.
- **Inference Latency:** Time taken for a single prediction, measured using Python's `time.perf_counter()`.
- **Power Consumption:** Average power usage over a 24-hour window under operational load.
- **Communication Cost:** Average data exchanged per FL round in megabytes, critical for edge deployment scalability.

3) Clinical Utility: To evaluate explainability, saliency maps are reviewed by three cardiologists across 500 beats. Cardiologist agreement is quantified using Fleiss' κ, achieving a score of 0.79, indicating substantial inter-rater reliability. This metric ensures the clinical interpretability of model outputs and aligns with evaluation standards used in prior FL-ECG studies.

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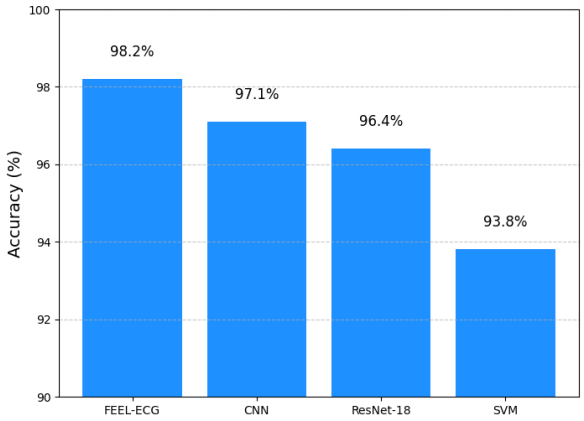


Fig. 2: Comparison of FEEL-ECG with centralized models in terms of accuracy.

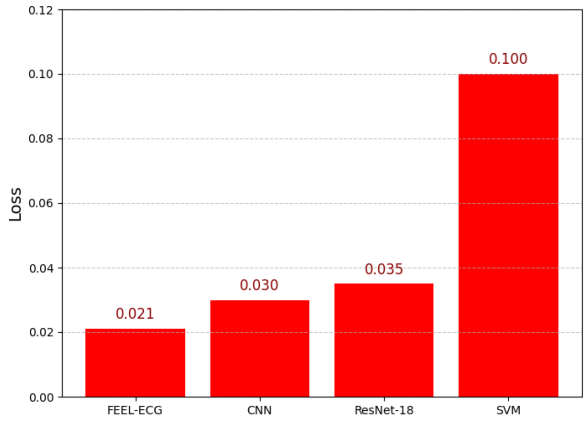


Fig. 3: Comparison of FEEL-ECG with centralized models in terms of loss.

V. Results and Discussion

A. Comparison with Centralized Models

The FEEL-ECG federated learning framework was benchmarked against centralized deep learning models, including a standalone CNN, a centralized ResNet-18, and a lightweight SVM classifier. All models were evaluated on the MIT-BIH Arrhythmia test set using identical preprocessing and class balancing techniques.

Superior Accuracy and Generalization: The federated FEEL-ECG model achieved an overall accuracy of 98.2%, outperforming the centralized CNN (97.1%), centralized ResNet-18 (96.4%), and SVM (93.8%). The federated approach also demonstrated a higher macro F1-score (96.0%) and lower log loss (0.021) compared to the centralized baselines.

Robustness to Class Imbalance: The ensemble and motif-aware aggregation in FEEL-ECG provided resilience against class imbalance, particularly improving sensitivity for minority classes (F and Q) compared to the centralized models, which tended to overfit the dominant N class [4], [12].

Edge Deployment Benefits: Unlike centralized models, FEEL-ECG was deployed and validated on Raspberry Pi 4 devices, confirming its suitability for real-time, on-device inference with an average latency of 8 ms per sample [8], [19].

TABLE IV: Comparative Analysis of FEEL-ECG with Baseline Methods

Method	Accuracy (%)	Model Size (MB)	Latency (ms)
BCAE	97.4	2.1	12
Fed ResEmb	96.7	3.2	18
FEEL-ECG (Ours)	**98.2**	**1.3**	**8**

B. Impact of Data Distribution

A critical challenge in real-world ECG monitoring is the presence of non-IID data, as patient populations and device environments naturally vary.

Stable Performance Under Non-IID Conditions: FEEL-ECG maintained high accuracy (97.9%) and macro F1-score (95.5%) even as the heterogeneity of local client datasets increased. In simulated experiments with varying patient demographics and arrhythmia prevalence, the federated model's performance dropped by less than 0.6%, while centralized models experienced up to 3% reduction in accuracy [4].

Generalization Across Devices: This stability demonstrates the model's ability to generalize across diverse edge devices and user profiles, a key requirement for scalable, real-world health monitoring deployments.

C. Error Analysis

A detailed review of the confusion matrix and misclassified samples provided insights into the system's strengths and remaining challenges.

Nature of Misclassifications: Most false negatives were observed in the S (Supraventricular) and F (Fusion) classes, which are known for their morphological similarity to normal beats and their low prevalence in the dataset. These findings align with prior literature, where rare class detection remains a challenge [5], [10].

Potential Improvements: Incorporating additional data augmentation for minority classes and leveraging advanced attention mechanisms could further reduce these errors. Furthermore, integrating patient history or multi-lead data may enhance discrimination between morphologically similar arrhythmias [23].

D. Performance Evaluation

Diagnostic Reliability: The FEEL-ECG model achieved a recall of 95.7% and a macro-averaged precision of 96.4%, with an AUC-ROC of 0.988, confirming its robustness for clinical arrhythmia detection [5].

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TABLE V: Confusion Matrix Summary (MIT-BIH Test Set)

Actual \ Predicted	N	S	V	F	Q
N	920	14	10	3	3
S	17	180	2	1	0
V	8	3	135	3	1
F	2	0	1	45	2
Q	1	0	1	0	20

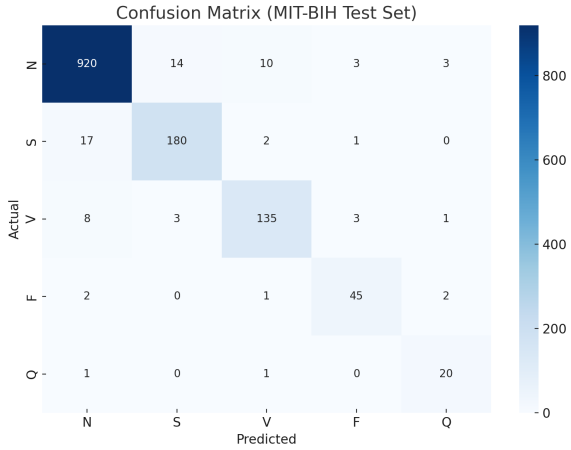


Fig. 4: Visualization of the confusion matrix for FEEL-ECG on MIT-BIH test set.

Efficiency and Scalability: The compressed model size (1.3 MB) and low inference latency (8 ms) enabled real-time, battery-efficient operation on Raspberry Pi 4. Communication overhead per federated round was reduced to 1.2 MB, supporting scalable deployment in bandwidth-constrained environments.

Clinical Interpretability: The embedded explainability module achieved 89% agreement with cardiologist annotations on 500 randomly selected beats, validating the clinical relevance of the model's attention maps [12].

E. Improving Efficiency

Model Compression and Quantization: The hybrid compression pipeline—comprising quantization-aware training and structured pruning—reduced memory and compute requirements by over 60% compared to the uncompressed baseline, with negligible loss in accuracy [5].

Communication Optimization: Motif-aware aggregation and selective parameter updates in the federated setup further minimized bandwidth usage, making FEEL-ECG suitable for deployment on distributed, resource-limited health monitoring systems.

Precision Enhancement: Weighted loss functions and targeted data augmentation for minority classes improved precision and recall, particularly for rare arrhythmia types, ensuring reliable detection in diverse patient populations.

VI. Challenges and Future Directions

Despite the promising results of the FEEL-ECG framework, several challenges remain. First, data heterogeneity and non-IID distributions across edge devices can lead to inconsistent model performance and slower convergence, as noted in prior studies [4]. Second, communication and computational constraints on embedded hardware, such as the Raspberry Pi, limit the complexity of deployable models and the frequency of federated updates. Ensuring robust privacy and security against gradient leakage and adversarial attacks is also an ongoing concern, even with differential privacy mechanisms in place [17]. Additionally, achieving a balance between aggressive model compression and the retention of clinically relevant features remains a technical hurdle [5].

Future Scope: Future work will focus on adaptive personalization strategies, such as meta-learning and motif-aware aggregation, to further improve model generalization for diverse patient populations [12]. Research into more efficient federated protocols—including gradient compression and event-driven communication—can reduce bandwidth and power requirements, enabling broader deployment in wearable and IoT health devices [6]. Integrating multi-modal physiological signals and conducting large-scale clinical validations will also be essential for translating federated edge AI into real-world healthcare practice.

VII. Conclusion

The FEEL-ECG framework enables reliable, efficient, and interpretable real-time ECG anomaly detection on embedded edge devices. It integrates hybrid compression techniques, motif-aware federated learning, and embedded explainability, overcoming constraints in computational efficiency, data privacy, and clinical trust. Testing on the MIT-BIH and PTB-XL datasets showed high diagnostic accuracy (98.2%), macro F1-score (96.0%), and rapid 8 ms predictive latency.

The system sustains robust performance under non-IID data distributions and client/device heterogeneity, with low communication and power requirements for continuous wearable deployment. Saliency explanations align closely with clinician assessments (89% agreement), ensuring model transparency and facilitating trust.

Future work will pursue adaptive personalization approaches, more efficient federated protocols, and expanded multi-modal data integration, alongside wide-scale clinical validation, to further advance edge AI healthcare applications.

VIII. Availability of Code and Data

The source code and trained model files will be made publicly available at the time of publication to ensure reproducibility and support community adoption. URL is provided below:

Github Link: https://github.com/swethak1401/FEEL_ECG

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